



University Institute of Engineering

Department of Computer Science & Engineering

COMPUTER ORGANIZATION & ARCHITECTURE

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BE-CSE

AUXILIARY MEMORY

- Secondary memory is a type of computer memory that is used for long-term storage of data and programs. It is also known as auxiliary memory or external memory, and is distinct from primary memory, which is used for short-term storage of data and instructions that are currently being processed by the CPU.
- Secondary memory devices are typically larger and slower than primary memory, but offer a much larger storage capacity. This makes them ideal for storing large files such as documents, images, videos, and other multimedia content.
- Primary memory has limited storage capacity and is volatile. Secondary memory overcomes this limitation by providing permanent storage of data and in bulk quantity.

AUXILIARY MEMORY

- Secondary memory is also termed external memory and refers to the various storage media on which a computer can store data and programs.
- The Secondary storage media can be fixed or removable. Fixed Storage media is an internal storage medium like a hard disk that is fixed inside the computer. A storage medium that is portable and can be taken outside the computer is termed removable storage media.
- Some examples of secondary memory devices include hard disk drives (HDDs), solid-state drives (SSDs), magnetic tapes, optical discs such as CDs and DVDs, and flash memory such as USB drives and memory cards. Each of these devices uses different technologies to store data, but they all share the common feature of being non-volatile, meaning that they can store data even when the computer is turned off.

AUXILIARY MEMORY

- Secondary memory devices are accessed by the CPU via input/output (I/O) operations, which involve transferring data between the device and primary memory.
- The speed of these operations is affected by factors such as the type of device, the size of the file being accessed, and the type of connection between the device and the computer.
- Overall, secondary memory is an essential component of modern computing systems and plays a critical role in the storage and retrieval of data and programs.

Difference between Primary Memory and Secondary Memory:

Primary Memory	Secondary Memory
Primary memory is directly accessed by the Central Processing Unit (CPU).	Secondary memory is not accessed directly by the Central Processing Unit (CPU). Instead, data accessed from a secondary memory is first loaded into Random Access Memory (RAM) and is then sent to the Processing Unit.
RAM provides a much faster-accessing speed to data than secondary memory. By loading software programs and required files into primary memory (RAM), computers can process data much more quickly.	Secondary Memory is slower in data accessing. Typically, primary memory is six times faster than secondary memory.

Primary Memory	Secondary Memory
Primary memory, i.e. Random Access Memory (RAM) is volatile and gets completely erased when a computer is shut down.	Secondary memory provides a feature of being non-volatile, which means it can hold on to its data with or without electrical power supply.

Uses of Secondary Storage:

- **Permanent Storage:** Primary Memory (RAM) is volatile, i.e. it loses all information when the electricity is turned off, so in order to secure the data permanently in the device, Secondary storage devices are needed.
- **Portability:** Storage mediums, like CDs, flash drives can be used to transfer the data from one device to another.

Fixed and Removable Storage Fixed Storage-

- Fixed storage is an internal media device that is used by a computer system to store data, and usually, these are referred to as the Fixed disk drives or Hard Drives.
- Fixed storage devices are literally not fixed, obviously, these can be removed from the system for repairing work, maintenance purposes, and also for an upgrade, etc.
- But in general, this can't be done without a proper toolkit to open up the computer system to provide physical access, and that needs to be done by an engineer.
- Technically, almost all of the data i.e. being processed on a computer system is stored on some type of a built-in fixed storage device.

Types of fixed storage:

- Internal flash memory (rare)
- SSD (solid-state disk) units
- Hard disk drives (HDD)

Removable Storage-

Removable storage is an external media device that is used by a computer system to store data, and usually, these are referred to as the Removable Disks drives or the External Drives. Removable storage is any type of storage device that can be removed/ejected from a computer system while the system is running. Examples of external devices include CDs, DVDs, and Blu-ray disk drives, as well as diskettes and USB drives. Removable storage makes it easier for a user to transfer data from one computer system to another. In storage factors, the main benefit of removable disks is that they can provide the fast data transfer rates associated with storage area networks (SANs)

Types of Removable Storage:

- Optical discs (CDs, DVDs, Blu-ray discs)
- Memory cards
- Floppy disks
- Magnetic tapes
- Disk packs
- Paper storage (punched tapes, punched cards)

Advantages:

- 1. Large storage capacity:** Secondary memory devices typically have a much larger storage capacity than primary memory, allowing users to store large amounts of data and programs.
- 2. Non-volatile storage:** Data stored on secondary memory devices is typically non-volatile, meaning it can be retained even when the computer is turned off.
- 3. Portability:** Many secondary memory devices are portable, making it easy to transfer data between computers or devices.
- 4. Cost-effective:** Secondary memory devices are generally more cost-effective than primary memory.

Disadvantages:

- 1. Slower access times:** Accessing data from secondary memory devices typically takes longer than accessing data from primary memory.
- 2. Mechanical failures:** Some types of secondary memory devices, such as hard disk drives, are prone to mechanical failures that can result in data loss.
- 3. Limited lifespan:** Secondary memory devices have a limited lifespan, and can only withstand a certain number of read and write cycles before they fail.
- 4. Data corruption:** Data stored on secondary memory devices can become corrupted due to factors such as electromagnetic interference, viruses, or physical damage.

Overall, secondary memory is an essential component of modern computing systems, but it also has its limitations and drawbacks. The choice of a particular secondary memory device depends on the user's specific needs and requirements.

References

Reference Books:

- J.P. Hayes, “Computer Architecture and Organization”, Third Edition.
- Mano, M., “Computer System Architecture”, Third Edition, Prentice Hall.
- Stallings, W., “Computer Organization and Architecture”, Eighth Edition, Pearson Education.

Text Books:

- Carpinelli J.D,” Computer systems organization &Architecture”, Fourth Edition, Addison Wesley.
- Patterson and Hennessy, “Computer Architecture”, Fifth Edition Morgan Kaufman.

Reference Website:

- [Memory Hierarchy Design and its Characteristics - GeeksforGeeks](#)
- [What is memory hierarchy? \(tutorialspoint.com\)](#)

Video Links:

- https://youtu.be/oj4_c9IMOCg?si=1FMiriyxX7JwkE4h
- <https://youtu.be/zwovvWfkuSg?si=fKl4vq0THSpiMa0M>
- <https://youtu.be/NVUWlO5zsk0?si=blZS6dXlR5NJOmKI>
- https://youtu.be/_dfsb7Gwems?si=zYh3TYajVcYHgAe9
- https://youtu.be/w04AMAEr60k?si=oWfL6Rrk0_Yf-gZ0